

AI Database Intelligence Report

AI Database Summary

The Olist database is a compact e-commerce data model that connects customers, orders, products, sellers, payments, reviews, and geographic information. With roughly 1 million rows of location data, 99 k customers and orders, 112 k order items, 33 k products, and 3 k sellers, it offers a rich, cross-functional view of the entire sales cycle. Companies can use it to segment shoppers, forecast demand, optimize shipping, and improve product and marketing strategies.

Key takeaways

- **Core entities** – Customers → Orders → Order Items → Products & Sellers, plus Payments and Reviews linked to Orders.
 - **Data scale** – ~1 M geolocation rows, ~99 k customers/orders, ~112 k items, ~33 k products, ~3 k sellers.
 - **Geographic linkage** – Zip-code prefixes tie customers, sellers, and geolocation data for regional analysis.
 - **Business uses** – Customer segmentation, sales performance, shipping logistics, payment behavior, and review sentiment analysis.
 - **Extensibility** – Category translations allow multi-language product catalog insights.
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Data Dictionary

olist_customers_dataset

Column	Type	Description
customer_id	object	Internal numeric ID that uniquely identifies each customer record in the database.
customer_unique_id	object	Alphanumeric code that uniquely identifies a customer across different systems or datasets.
customer_zip_code_prefix	int64	The first few digits of the customer's ZIP/postal code, used to group customers by geographic region.
customer_city	object	The city where the customer's address is located.
customer_state	object	The state or province in which the customer's address is located.

olist_geolocation_dataset

Column	Type	Description
geolocation_zip_code_prefix	int64	The first few digits of the ZIP code that identify the general area of the location.
geolocation_lat	float64	The latitude coordinate that pinpoints the location's north-south position on the globe.
geolocation_lng	float64	The longitude coordinate that pinpoints the location's east-west position on the globe.
geolocation_city	object	The city where the location is situated.
geolocation_state	object	The state or province where the location is situated.

olist_order_items_dataset

Column	Type	Description
order_id	object	
order_item_id	int64	
product_id	object	
seller_id	object	
shipping_limit_date	object	
price	float64	
freight_value	float64	

olist_order_payments_dataset

Column	Type	Description
order_id	object	unique identifier for each order
payment_sequential	int64	sequence number of the payment installment
payment_type	object	method of payment used (e.g., credit card, boleto)
payment_installments	int64	total number of installments for the order
payment_value	float64	amount paid in this installment

olist_order_reviews_dataset

Column	Type	Description
review_id	object	
order_id	object	
review_score	int64	
review_comment_title	object	
review_comment_message	object	
review_creation_date	object	
review_answer_timestamp	object	

olist_orders_dataset

Column	Type	Description
order_id	object	unique identifier for each order
customer_id	object	unique identifier for the customer who placed the order
order_status	object	current state of the order (e.g., pending, approved, shipped, delivered)
order_purchase_timestamp	object	date and time the customer placed the order
order_approved_at	object	date and time the seller approved the order for processing
order_delivered_carrier_date	object	date and time the carrier shipped the order
order_delivered_customer_date	object	date and time the customer actually received the order
order_estimated_delivery_date	object	projected date the order should arrive at the customer's address

olist_products_dataset

Column	Type	Description
product_id	object	Unique identifier for each product
product_category_name	object	Category or type of the product (e.g., “Electronics”, “Home & Kitchen”)
product_name_lenght	float64	Number of characters in the product’s name
product_description_lenght	float64	Number of characters in the product’s description
product_photos_qty	float64	How many photos are available for the product
product_weight_g	float64	Weight of the product in grams
product_length_cm	float64	Length dimension of the product in centimeters
product_height_cm	float64	Height dimension of the product in centimeters
product_width_cm	float64	Width dimension of the product in centimeters

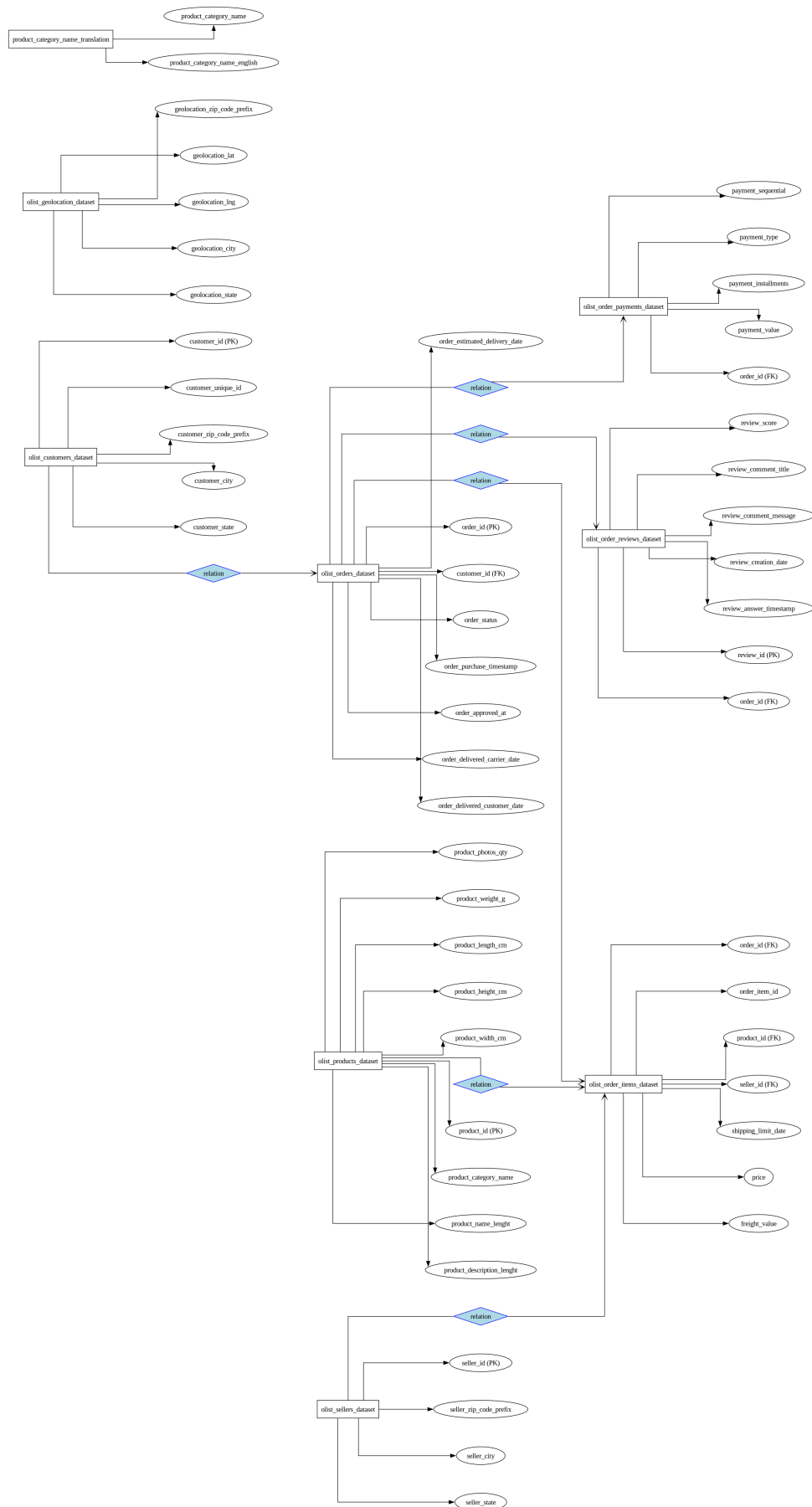
olist_sellers_dataset

Column	Type	Description
seller_id	object	Unique identifier for each seller in the marketplace.
seller_zip_code_prefix	int64	The first part of the seller’s ZIP code, indicating the general geographic area.
seller_city	object	The city where the seller’s business is located.
seller_state	object	The state where the seller’s business is located.

product_category_name_translation

Column	Type	Description
product_category_name	object	The name of the product category in the local language.
product_category_name_english	object	The same category name translated into English.

ER Diagram



Database Schema

olist_customers_dataset (99441 rows)

Column	Type	Constraint	Missing	Unique
customer_id	object	PK	0	99441
customer_unique_id	object		0	96096
customer_zip_code_prefix	int64		0	14994
customer_city	object		0	4119
customer_state	object		0	27

olist_geolocation_dataset (1000163 rows)

Column	Type	Constraint	Missing	Unique
geolocation_zip_code_prefix	int64		0	19015
geolocation_lat	float64		0	717360
geolocation_lng	float64		0	717613
geolocation_city	object		0	8011
geolocation_state	object		0	27

olist_order_items_dataset (112650 rows)

Column	Type	Constraint	Missing	Unique
order_id	object	FK	0	98666
order_item_id	int64		0	21
product_id	object	FK	0	32951
seller_id	object	FK	0	3095
shipping_limit_date	object		0	93318
price	float64		0	5968
freight_value	float64		0	6999

olist_order_payments_dataset (103886 rows)

Column	Type	Constraint	Missing	Unique
order_id	object	FK	0	99440
payment_sequential	int64		0	29
payment_type	object		0	5
payment_installments	int64		0	24
payment_value	float64		0	29077

olist_order_reviews_dataset (99224 rows)

Column	Type	Constraint	Missing	Unique
review_id	object	PK	0	98410
order_id	object	FK	0	98673
review_score	int64		0	5
review_comment_title	object		87656	4527
review_comment_message	object		58247	36159
review_creation_date	object		0	636
review_answer_timestamp	object		0	98248

olist_orders_dataset (99441 rows)

Column	Type	Constraint	Missing	Unique
order_id	object	PK	0	99441
customer_id	object	FK	0	99441
order_status	object		0	8
order_purchase_timestamp	object		0	98875
order_approved_at	object		160	90733
order_delivered_carrier_date	object		1783	81018
order_delivered_customer_date	object		2965	95664
order_estimated_delivery_date	object		0	459

olist_products_dataset (32951 rows)

Column	Type	Constraint	Missing	Unique
product_id	object	PK	0	32951
product_category_name	object		610	73
product_name_lenght	float64		610	66
product_description_lenght	float64		610	2960
product_photos_qty	float64		610	19
product_weight_g	float64		2	2204
product_length_cm	float64		2	99
product_height_cm	float64		2	102
product_width_cm	float64		2	95

olist_sellers_dataset (3095 rows)

Column	Type	Constraint	Missing	Unique
seller_id	object	PK	0	3095
seller_zip_code_prefix	int64		0	2246
seller_city	object		0	611
seller_state	object		0	23

product_category_name_translation (71 rows)

Column	Type	Constraint	Missing	Unique
product_category_name	object		0	71
product_category_name_english	object		0	71

Data Quality Analysis

Table	Rows	Cols	Completeness %	Duplicates	Dup Rate %	Null-heavy	Memory MB
olist_customers_dataset	99441	5	100.0	0	0.0	0	27.88
olist_geolocation_dataset	1000163	5	100.0	261831	26.18	0	136.59
olist_order_items_dataset	112650	7	100.0	0	0.0	0	37.74
olist_order_payments_dataset	103886	5	100.0	0	0.0	0	17.02
olist_order_reviews_dataset	99224	7	78.99	0	0.0	2	41.03
olist_orders_dataset	99441	8	99.38	0	0.0	0	55.51
olist_products_dataset	32951	9	99.17	0	0.0	0	6.6
olist_sellers_dataset	3095	4	100.0	0	0.0	0	0.62
product_category_name_translation	71	2	100.0	0	0.0	0	0.01

Business Insights (AI Generated)

- **Customer base:** 99 441 unique customers, no duplicates – strong, clean customer data.
- **Geolocation data:** 1 000 163 rows with 261 831 duplicate records – likely many customers share the same location, but duplicates could inflate location counts and skew regional analysis.
- **Orders vs. Items:** 99 441 orders match 112 650 order items – average 1.13 items per order, indicating most orders are single-product purchases.
- **Payments:** 103 886 payment records for 99 441 orders – some orders have multiple payments or missing payment entries, suggesting possible split-payment or refund scenarios.
- **Reviews:** 99 224 reviews but only 78.99 % completeness – missing review data for ~20 % of orders, which could bias satisfaction metrics.
- **Products:** 32 951 products with 99.17 % completeness – a few products have missing attributes, potentially affecting inventory and recommendation accuracy.
- **Sellers:** 3 095 sellers – relatively small seller pool, but no duplicates, indicating reliable seller identification.
- **Category translation:** 71 category names – useful for multilingual reporting, but ensure translations are consistently applied across product data.

Anomalies / Data quality issues

- Duplicate geolocation entries may inflate regional sales figures.
- Missing review data (~20 %) could under-represent customer sentiment.
- Incomplete product attributes (0.83 %) may lead to inaccurate product performance analysis.

Relationships / Correlations

- Each order links to one customer, one seller, and one or more items.
- Payment records are tied to orders; discrepancies may reveal payment issues or refunds.
- Reviews are tied to orders; missing reviews correlate with incomplete order data.

Opportunities

- Clean up geolocation duplicates to improve accurate regional marketing.

- Target customers with missing reviews for follow-up surveys to boost data completeness.
- Leverage the high single-item order rate to promote bundle deals and upsells.

Risks

- Duplicate geolocations could mislead inventory stocking decisions.
- Incomplete reviews and product data may lead to misinformed business decisions.
- Multiple payments per order could indicate fraud or payment processing issues that need monitoring.